

A counterexample to the minimum order clause in TLMC 00000008420

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Abstract

The conjecture includes the assertion that the smallest order of a Kirkman triple system admitting a transitive automorphism is 15. Under its stated definition, the one-block system of order 3 is a counterexample. Its three-cycle is a single transitive automorphism, including preservation of the resolution. Thus the conjecture, as a conjunction, is false.

1 The clause and its definitions

The English statement defines a Kirkman triple system as a resolvable Steiner triple system and claims that “the smallest order of a KTS with a transitive automorphism is the point-transitive type of order 15.” We use the stronger literal interpretation: a single automorphism must generate a point-transitive cyclic action.

A Steiner triple system on a finite set V has a collection $\mathcal{B}$ of three-element subsets such that each pair of distinct points belongs to exactly one block. A resolution is a partition $\mathcal{R}$ of $\mathcal{B}$ into parallel classes, each of which partitions V . An automorphism of the resolved system is a permutation of V that preserves the blocks and the resolution. The statement does not impose $n > 3$ or require a nontrivial system.

2 The counterexample

Proposition 1. *There exists a Kirkman triple system of order 3 with a single transitive automorphism. Consequently the asserted minimum order 15 is false.*

Proof. Set

$$V = \{0, 1, 2\}, \quad B = \{0, 1, 2\}, \quad \mathcal{B} = \{B\}, \quad \mathcal{R} = \{\{B\}\}.$$

The only block has three distinct points. Each of the three unordered pairs of distinct points lies in B , and there is no second block. Hence $(V, \mathcal{B})$ is a Steiner triple system. The sole parallel class $\{B\}$ partitions V , and the collection containing that class partitions $\mathcal{B}$. This is a Kirkman triple system.

Let $\sigma = (0\ 1\ 2)$. It is a bijection, fixes B setwise, and fixes the sole parallel class. It is therefore an automorphism of the resolved system. Its iterates take any given point to all three points, so the cyclic group generated by this single automorphism acts transitively. In particular, σ has order exactly 3.

The claimed minimum would imply that every such system has order at least 15. The constructed system has order $3 < 15$, a contradiction. $\square$

3 Scope and the earlier rejected argument

This argument refutes the minimum-order conjunct. It makes no claim against the existence criterion or the formula for the number of parallel classes. Indeed, the example has $(3-1)/2 = 1$ parallel class. It also does not refute the separate assertion about a full automorphism group of prime cyclic order: the full automorphism group of this example is S_3 , not C_3 . The relevant fact is that it *contains* the transitive three-cycle.

The earlier submission in pull request 21 used the order-9 affine plane and a transitive translation group. The subsequent audit removed that submission because a transitive group does not establish the existence of a single transitive automorphism. A nonzero translation of $\mathbb{F}_3^2$ has order 3 and three orbits of size 3, so it cannot be transitive on nine points. The audit additionally observes that $\text{AGL}(2, 3)$ has no element of order 9. The present counterexample avoids that substitution by displaying the required single permutation explicitly. It addresses the statement as written; an amended statement excluding order 3 would require a different analysis.

The order-3 convention is standard, not an invented relaxation. Brown and Mellinger’s discussion explicitly includes a KTS with three points, one block, and one parallel class, while distinguishing the smallest nontrivial case of order 9 [2, PDF page 3].

4 Formalization

The accompanying `Counterexample.lean` imports only `Std`. Its finite incidence model separately indexes points, blocks, and resolution classes. It states and checks: exactly three distinct points in each block; unique containment of each pair; unique containment of each point within each resolution class; bijectivity on all three index sets; preservation of incidence and resolution; and transitivity under iterates of the same point permutation. It also checks that this permutation has exact order 3.

The theorem `minimum_order_clause_false` negates a minimum-order assertion with both attainment at 15 and a lower-bound condition. The final conjunction theorem records why falsifying this conjunct suffices, without purporting to prove or disprove the other conjuncts. There are no admitted proofs or added axioms in the source.

References

- [1] The Last Math Competition, *Conjecture 00000008420*, repository statement. Repository statement.
- [2] E. Brown and K. E. Mellinger, *Kirkman’s Schoolgirls Wearing Hats and Walking through Fields of Numbers*, Mathematics Magazine 82 (2009), no. 1. <https://personal.math.vt.edu/brown/doc/kirkman.pdf>.
- [3] The Last Math Competition, pull request 21 and subsequent audit audit commit 541cf4f. Earlier pull request.